

OpenStax Calculus Volume 3

Section 6.8 Exercises: The Divergence Theorem

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For the following exercises, use a computer algebraic system (CAS) and the divergence theorem to evaluate surface integral $\iint_S \mathbf{F} \cdot \mathbf{N} ds$ for the given choice of $\mathbf{F}$ and the boundary surface S . For each closed surface, assume $\mathbf{N}$ is the outward unit normal vector.

1. [T] $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$; S is the surface of cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$.
2. [T] $\mathbf{F}(x, y, z) = (\cos yz)\mathbf{i} + e^{xz}\mathbf{j} + 3z^2\mathbf{k}$; S is the surface of hemisphere $z = \sqrt{4 - x^2 - y^2}$ together with disk $x^2 + y^2 \leq 4$ in the xy -plane.
3. [T] $\mathbf{F}(x, y, z) = (x^2 + y^2 - x^2)\mathbf{i} + x^2y\mathbf{j} + 3z\mathbf{k}$; S is the surface of the unit cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$ excluding the face $z = 0$.
4. [T] $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$; S is the surface of the solid bounded by the parabola $z = x^2 + y^2$ and the plane $z = 9$.
5. [T] $\mathbf{F}(x, y, z) = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$; S is the surface of sphere $x^2 + y^2 + z^2 = 4$.
6. [T] $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + (z^2 - 1)\mathbf{k}$; S is the surface of the solid bounded by cylinder $x^2 + y^2 = 4$ and planes $z = 0$ and $z = 1$.
7. [T] $\mathbf{F}(x, y, z) = xy^2\mathbf{i} + yz^2\mathbf{j} + x^2z\mathbf{k}$; S is the surface of the solid bounded above by sphere $\rho = 2$ and below by cone $\phi = \frac{\pi}{4}$ in spherical coordinates. (Think of S as the surface of an “ice cream cone.”)
8. [T] $\mathbf{F}(x, y, z) = x^3\mathbf{i} + y^3\mathbf{j} + 3a^2z\mathbf{k}$ (constant $a > 0$); S is the surface of the solid bounded by cylinder $x^2 + y^2 = a^2$ and planes $z = 0$ and $z = 1$.
9. [T] Surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where S is the surface of the solid bounded by paraboloid $z = x^2 + y^2$ and plane $z = 4$, and $\mathbf{F}(x, y, z) = (x + y^2z^2)\mathbf{i} + (y + z^2x^2)\mathbf{j} + (z + x^2y^2)\mathbf{k}$
10. Use the divergence theorem to calculate surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = (e^{y^2})\mathbf{i} + (y + \sin(z^2))\mathbf{j} + (z - 1)\mathbf{k}$ and S is the surface of the solid bounded by the sphere $x^2 + y^2 + z^2 = 1$, and below by the plane $z = 0$.
11. Use the divergence theorem to calculate surface integral $\iint_S \mathbf{F} \cdot d\mathbf{s}$, where $\mathbf{F}(x, y, z) = x^4\mathbf{i} - x^3z^2\mathbf{j} + 4xy^2z\mathbf{k}$ and S is the surface bounded by cylinder $x^2 + y^2 = 1$ and planes $z = x + 2$ and $z = 0$.
12. Use the divergence theorem to calculate surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ when $\mathbf{F}(x, y, z) = x^2z^3\mathbf{i} + 2xyz^3\mathbf{j} + xz^4\mathbf{k}$ and S is the surface of the box with vertices $(\pm 1, \pm 2, \pm 3)$.

13. Use the divergence theorem to calculate surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ when $\mathbf{F}(x, y, z) = z \tan^{-1}(y^2)\mathbf{i} + z^3 \ln(x^2 + 1)\mathbf{j} + z\mathbf{k}$ and S is the surface of the solid bounded by the paraboloid $x^2 + y^2 + z = 2$ and the plane $z = 1$.
14. [T] Use a CAS and the divergence theorem to calculate flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = (x^3 + y^3)\mathbf{i} + (y^3 + z^3)\mathbf{j} + (z^3 + x^3)\mathbf{k}$ and S is a sphere with center $(0, 0, 0)$ and radius 2.
15. Use the divergence theorem to compute the value of flux integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = (y^3 + 3x)\mathbf{i} + (xz + y)\mathbf{j} + [z + x^4 \cos(x^2 y)]\mathbf{k}$ and S is the surface of the solid bounded by $x^2 + y^2 = 1, x \geq 0, y \geq 0$, and $0 \leq z \leq 1$.
[Figure: A vector field in three dimensions, with focus on the area with $x > 0, y > 0$, and $z > 0$. A quarter of a cylinder is drawn with center on the z axis. The arrows have positive x, y , and z components; they point away from the origin.]
16. Use the divergence theorem to compute flux integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = y\mathbf{j} - z\mathbf{k}$ and S consists of the union of paraboloid $y = x^2 + z^2, 0 \leq y \leq 1$, and disk $x^2 + z^2 \leq 1, y = 1$, oriented outward. What is the flux through just the paraboloid?
17. Use the divergence theorem to compute flux integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = x + y\mathbf{j} + z^4\mathbf{k}$ and S is a part of cone $z = \sqrt{x^2 + y^2}$ beneath top plane $z = 1$, oriented downward.
18. Use the divergence theorem to calculate surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ for $\mathbf{F}(x, y, z) = x^4\mathbf{i} - x^3 z^2\mathbf{j} + 4xy^2 z\mathbf{k}$, where S is the surface inside the cylinder $x^2 + y^2 = 1$ between the planes $z = x + 2$ and $z = 0$.
19. Consider $\mathbf{F}(x, y, z) = x^2\mathbf{i} + xy\mathbf{j} + (z + 1)\mathbf{k}$. Let E be the solid enclosed by paraboloid $z = 4 - x^2 - y^2$ and plane $z = 0$ with normal vectors pointing outside E . Compute flux $\mathbf{F}$ across the boundary of E using the divergence theorem.

For the following exercises, use a CAS along with the divergence theorem to compute the net outward flux for the fields across the given surfaces S .

20. [T] $\mathbf{F} = \langle x, -2y, 3z \rangle$; S is sphere $\{(x, y, z) : x^2 + y^2 + z^2 = 6\}$.
21. [T] $\mathbf{F} = \langle x, 2y, z \rangle$; S is the boundary of the tetrahedron in the first octant formed by plane $x + y + z = 1$.
22. [T] $\mathbf{F} = \langle y - 2x, x^3 - y, y^2 - z \rangle$; S is sphere $\{(x, y, z) : x^2 + y^2 + z^2 = 4\}$.
23. [T] $\mathbf{F} = \langle x, y, z \rangle$; S is the surface of paraboloid $z = 4 - x^2 - y^2$, for $z \geq 0$, plus its base in the xy -plane.

For the following exercises, use a CAS and the divergence theorem to compute the net outward flux for the vector fields across the boundary of the given regions D .

24. [T] $\mathbf{F} = \langle z - x, x - y, 2y - z \rangle$; D is the region between spheres of radius 2 and 4 centered at the origin.
25. [T] $\mathbf{F} = \frac{\mathbf{r}}{\|\mathbf{r}\|} = \frac{\langle x, y, z \rangle}{\sqrt{x^2 + y^2 + z^2}}$; D is the region between spheres of radius 1 and 2 centered at the origin.
26. [T] $\mathbf{F} = \langle x^2, -y^2, z^2 \rangle$; D is the region in the first octant between planes $z = 4 - x - y$ and $z = 2 - x - y$.

27. Let $\mathbf{F}(x, y, z) = 2x\mathbf{i} - 3xy\mathbf{j} + xz^2\mathbf{k}$. Use the divergence theorem to calculate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where S is the surface of the cube with corners at $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(1, 1, 0)$, $(0, 0, 1)$, $(1, 0, 1)$, $(0, 1, 1)$, and $(1, 1, 1)$, oriented outward.
28. Use the divergence theorem to find the outward flux of field $\mathbf{F}(x, y, z) = (x^3 - 3y)\mathbf{i} + (2yz + 1)\mathbf{j} + xyz\mathbf{k}$ through the cube bounded by planes $x = \pm 1$, $y = \pm 1$, and $z = \pm 1$.
29. Let $\mathbf{F}(x, y, z) = 2x\mathbf{i} - 3y\mathbf{j} + 5z\mathbf{k}$ and let S be hemisphere $z = \sqrt{9 - x^2 - y^2}$ together with disk $x^2 + y^2 \leq 9$ in the xy -plane. Use the divergence theorem to calculate $\iint_S \mathbf{F} \cdot d\mathbf{S}$.
30. Evaluate $\iint_S \mathbf{F} \cdot \mathbf{N} dS$, where $\mathbf{F}(x, y, z) = x^2\mathbf{i} + xy\mathbf{j} + x^3y^3\mathbf{k}$ and S is the surface consisting of all faces of the tetrahedron bounded by plane $x + y + z = 1$ and the coordinate planes, with outward unit normal vector $\mathbf{N}$.
[Figure: A vector field in three dimensions, with arrows becoming larger the further away from the origin they are, especially in their x components. S is the surface consisting of all faces of the tetrahedron bounded by the plane $x + y + z = 1$. As such, a portion of the given plane, the (x, y) plane, the (x, z) plane, and the (y, z) plane are shown. The arrows point towards the origin for negative x components, away from the origin for positive x components, down for positive x and negative y components, as well as positive y and negative x components, and for positive x and y components, as well as negative x and negative y components.]
31. Find the net outward flux of field $\mathbf{F} = \langle bz - cy, cx - az, ay - bx \rangle$ across any smooth closed surface in $\mathbf{R}^3$, where a , b , and c are constants.
32. Use the divergence theorem to evaluate $\iint_S \|\mathbf{R}\| \mathbf{R} \cdot \mathbf{n} dS$, where $\mathbf{R}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and S is sphere $x^2 + y^2 + z^2 = a^2$, with constant $a > 0$.
33. Use the divergence theorem to evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = y^2z\mathbf{i} + y^3\mathbf{j} + xz\mathbf{k}$ and S is the boundary of the cube defined by $-1 \leq x \leq 1$, $-1 \leq y \leq 1$, and $0 \leq z \leq 2$.
34. Let R be the region defined by $x^2 + y^2 + z^2 \leq 1$. Use the divergence theorem to find $\iiint_R z^2 dV$.
35. Let E be the solid bounded by the xy -plane and paraboloid $z = 4 - x^2 - y^2$ so that S is the surface of the paraboloid piece together with the disk in the xy -plane that forms its bottom. If $\mathbf{F}(x, y, z) = (xz \sin(yz) + x^3)\mathbf{i} + \cos(yz)\mathbf{j} + (3zy^2 - e^{x^2+y^2})\mathbf{k}$, find $\iint_S \mathbf{F} \cdot d\mathbf{S}$ using the divergence theorem.
[Figure: A vector field in three dimensions with all of the arrows pointing down. They seem to follow the path of the paraboloid drawn opening down with vertex at the origin. S is the surface of this paraboloid and the disk in the (x, y) plane that forms its bottom.]
36. Let E be the solid unit cube with diagonally opposite corners at the origin and $(1, 1, 1)$, and faces parallel to the coordinate planes. Let S be the surface of E , oriented with the outward-pointing normal. Use a CAS to find $\iint_S \mathbf{F} \cdot d\mathbf{S}$ using the divergence theorem if $\mathbf{F}(x, y, z) = 2xy\mathbf{i} + 3ye^z\mathbf{j} + x \sin z\mathbf{k}$.
37. Use the divergence theorem to calculate the flux of $\mathbf{F}(x, y, z) = x^3\mathbf{i} + y^3\mathbf{j} + z^3\mathbf{k}$ through sphere $x^2 + y^2 + z^2 = 1$.
38. Find $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and S is the outwardly oriented surface obtained by removing cube $[1, 2] \times [1, 2] \times [1, 2]$ from cube $[0, 2] \times [0, 2] \times [0, 2]$.
39. Consider radial vector field $\mathbf{F} = \frac{\mathbf{r}}{|\mathbf{r}|} = \frac{\langle x, y, z \rangle}{(x^2 + y^2 + z^2)^{1/2}}$. Compute the surface integral, where S is the surface of a sphere of radius a centered at the origin.

40. Compute the flux of water through parabolic cylinder $S : y = x^2$, from $0 \leq x \leq 2, 0 \leq z \leq 3$, if the velocity vector is $\mathbf{F}(x, y, z) = 3z^2\mathbf{i} + 6\mathbf{j} + 6xz\mathbf{k}$.
41. [T] Use a CAS to find the flux of vector field $\mathbf{F}(x, y, z) = z\mathbf{i} + z\mathbf{j} + \sqrt{x^2 + y^2}\mathbf{k}$ across the portion of hyperboloid $x^2 + y^2 = z^2 + 1$ between planes $z = 0$ and $z = \frac{\sqrt{3}}{3}$, oriented so the unit normal vector points away from the z-axis.
42. [T] Use a CAS to find the flux of vector field $\mathbf{F}(x, y, z) = (e^y + x)\mathbf{i} + (3\cos(xz) - y)\mathbf{j} + z\mathbf{k}$ through surface S, where S is given by $z^2 = 4x^2 + 4y^2$ from $0 \leq z \leq 4$, oriented so the unit normal vector points downward.
43. [T] Use a CAS to compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + 2z\mathbf{k}$ and S is the boundary of a part of solid sphere $x^2 + y^2 + z^2 \leq 2$ with $0 \leq z \leq 1$.
44. Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = bxy^2\mathbf{i} + bx^2y\mathbf{j} + (x^2 + y^2)z^2\mathbf{k}$ and S is the boundary of the solid cylinder $x^2 + y^2 \leq a^2$ and $0 \leq z \leq b$.
45. [T] Use a CAS to calculate the flux of $\mathbf{F}(x, y, z) = (x^3 + y \sin z)\mathbf{i} + (y^3 + z \sin x)\mathbf{j} + 3z\mathbf{k}$ across surface S, where S is the boundary of the solid bounded by hemispheres $z = \sqrt{4 - x^2 - y^2}$ and $z = \sqrt{1 - x^2 - y^2}$, and plane $z = 0$.
46. Use the divergence theorem to evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = xy\mathbf{i} - \frac{1}{2}y^2\mathbf{j} + z\mathbf{k}$ and S is the surface consisting of three pieces: $z = 4 - 3x^2 - 3y^2, 1 \leq z \leq 4$ on the top; $x^2 + y^2 = 1, 0 \leq z \leq 1$ on the sides; and $z = 0$ on the bottom.
47. [T] Use a CAS and the divergence theorem to evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = (2x + y \cos z)\mathbf{i} + (x^2 - y)\mathbf{j} + y^2z\mathbf{k}$ and S is sphere $x^2 + y^2 + z^2 = 4$ orientated outward.
48. Use the divergence theorem to evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and S is the boundary of the solid enclosed by paraboloid $y = x^2 + z^2 - 2$, cylinder $x^2 + z^2 = 1$, and plane $x + y = 2$, and S is oriented outward.

For the following exercises, Fourier's law of heat transfer states that the heat flow vector $\mathbf{F}$ at a point is proportional to the negative gradient of the temperature; that is, $\mathbf{F} = -k\nabla T$, which means that heat energy flows hot regions to cold regions. The constant $k > 0$ is called the conductivity, which has metric units of joules per meter per second-kelvin or watts per meter-kelvin. A temperature function for region D is given. Use the divergence theorem to find net outward heat flux $\iint_S \mathbf{F} \cdot \mathbf{N} dS = -k \iint_S \nabla T \cdot \mathbf{N} dS$ across the boundary S of D, where $k = 1$.

49. $T(x, y, z) = 100 + x + 2y + z$; $D = \{(x, y, z) : 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1\}$
50. $T(x, y, z) = 100 + e^{-z}$; $D = \{(x, y, z) : 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1\}$
51. $T(x, y, z) = 100e^{-x^2 - y^2 - z^2}$; D is the sphere of radius a centered at the origin.